

Project 12: AI-Discovered Tandem PV Pipeline

A standalone investor brief on the highest-scoring project (37/40) from the SciMSPT
Frontier Series Volume 02, covering technology, SciMSPT integration, financial
CLEAR WINNER · 37 / 40 in 18-month capital-deployment roadmap.

\$480B

2040 TAM

34%

10YR IRR

4.9yr

PAYBACK

3

GTM PHASES

Table of Contents

Table of Contents	1
1. Executive Summary: Why P12 Wins	2
2. Technology Deep Dive	3
2.1 Tandem Perovskite-Silicon Architecture	3
2.2 The AI Discovery Layer	3
2.3 Source Breakthroughs (2025-2026)	3
3. SciMSPT Platform Integration	5
3.1 Workload Profile	5
3.2 Multi-Tenancy and Data Isolation	5
4. Multi-Factor Scorecard (37/40)	6
4.1 Interpreting the Score	6
5. Market Sizing: \$70B to \$480B	7
5.1 Serviceable Addressable Market	7
5.2 Competitive Landscape	7
6. Financial Model & Scenario Analysis	8
6.1 Capital Ask by Phase	8
6.2 Sensitivity to Module ASP	8
7. Three-Phase Go-to-Market Strategy	9
7.1 Beachhead Customer Selection	9
7.2 Pricing Model	9
8. 18-Month Capital-Deployment Roadmap	10
8.1 Capital Efficiency Rationale	10
9. Risk Register & Mitigation	11
9.1 The Three Risks That Could Kill the Company	11
10. Strategic Recommendation & Capital Ask	13
10.1 Why \$20M Now	13
10.2 Why Reserve \$80M for Series B	13
10.3 What the Investor Gets	13
11. References	14

1. Executive Summary: Why P12 Wins

37/40 Score	\$480B 2040 TAM	34% 10yr IRR	4.9yr Payback	60% Cross-domain leverage
-----------------------	---------------------------	------------------------	-------------------------	-------------------------------------

Project 12 — the AI-Discovered Tandem PV Pipeline — is the highest-scoring project across all twelve ventures profiled in *Beyond the Energy Trilemma* (Vol. 02 of the SciMSPT Foresight Series). It scored 37 out of a possible 40 on an eight-factor machine-learning scoring matrix that weighed technology readiness, total addressable market, internal rate of return, cross-domain leverage, policy tailwind, competitive moat, SciMSPT platform fit, and capital efficiency. No other project in either volume of the Foresight Series exceeded 33 points. P12 is, by a clear margin, the recommended lead investment for any foresight-driven capital allocator constructing a clean-energy portfolio.

The thesis rests on five compounding arguments. **First**, the project sits on a capital-light discovery layer — graphics-processing-unit hours and a proprietary training corpus — that can be monetised as software-as-a-service before any gigafactory capital is committed. **Second**, the underlying physical asset (tandem perovskite-silicon photovoltaics) has the largest total addressable market in the entire Matter & Energy portfolio: roughly \$70 billion in 2026, growing to an estimated \$480 billion by 2040 as the world adds 700-900 gigawatts of solar capacity per year. **Third**, the AI discovery layer is a self-reinforcing data moat: every formulation screened trains the next discovery run, producing a network effect that is structurally unavailable to traditional empirical materials discovery. **Fourth**, the project enjoys direct policy tailwinds from the United States Inflation Reduction Act §48E investment tax credit, the European Union Green Deal, China's 2060 carbon-neutrality commitment, and India's Production-Linked Incentive scheme for high-efficiency solar manufacturing. **Fifth**, cheaper clean electricity directly enables at least three other projects in the portfolio — solid-state battery gigafactories (P3), green hydrogen electrolyser (P5), and direct air capture (P11) — producing a 60% cross-domain leverage factor that no other project in the portfolio can match.

This brief unpacks the technology, SciMSPT microservices integration, financial model, three-phase go-to-market plan, risk register, and an 18-month capital-deployment roadmap. It is intended as a standalone document for investors evaluating a Series A commitment of \$15-25 million in the fourth quarter of 2026, or as supporting material for a Series B raise of \$80-120 million in the first quarter of 2029 to fund the joint-venture pilot line.

"Tandem perovskite-silicon photovoltaics, supercharged by an AI discovery layer, is the single highest-leverage capital allocation in the SciMSPT Foresight Series. The economics, policy tailwinds, and platform fit are mutually reinforcing — and the data moat compounds with every screening run."

2. Technology Deep Dive

2.1 Tandem Perovskite-Silicon Architecture

A tandem photovoltaic cell stacks a wide-bandgap top cell (typically a mixed-halide perovskite such as $\text{Cs}_{0.05}(\text{FA}_{0.83}\text{MA}_{0.17})_{0.95}\text{Pb}(\text{I}_{0.83}\text{Br}_{0.17})_3$) on top of a narrow-bandgap bottom cell (silicon or, in next-generation stacks, tin-lead perovskite). The top cell harvests high-energy blue and ultraviolet photons that silicon cannot convert efficiently, while the bottom cell captures the remaining red and near-infrared spectrum. The theoretical Shockley-Queisser limit for a single-junction silicon cell is 33.7%; for a two-terminal tandem it rises to roughly 45%, and for a four-terminal tandem to 48%. Certified lab-cell efficiencies have already crossed 34% (Fraunhofer ISE, June 2026), and pilot-line module efficiencies above 22% are now reproducible across multiple manufacturers including Oxford PV, Longi, and Microquanta.

The performance gain alone — a 25-35% energy yield uplift over single-junction silicon at the module level — is sufficient to justify the additional manufacturing complexity. But the more strategically important property is that the perovskite top cell is solution-processable: it can be deposited by slot-die coating, blade coating, or vacuum thermal evaporation on top of a finished silicon heterojunction bottom cell. This means existing silicon gigafactories — which represent roughly \$400 billion of sunk capital worldwide — can be retrofitted to tandem production with a 10-15% incremental capex spend, rather than being stranded by a discontinuous technology transition.

2.2 The AI Discovery Layer

Where P12 diverges from every other tandem PV manufacturer in the market is in the AI discovery layer that selects the perovskite formulation. Rather than empirically testing compositions one at a time — a process that took the field fifteen years to converge on the current Cs-FA-MA-Pb-I-Br standard — P12 trains a graph neural network on a corpus of approximately 50,000 published and proprietary perovskite characterisation records. The model predicts four critical properties simultaneously: bandgap (target 1.68-1.73 eV), thermal stability (target > 85°C), moisture resistance (target < 1% degradation at 85% relative humidity over 1000 hours), and defect tolerance (target SRH lifetime > 100 ns).

The discovery layer is a closed loop with the wet lab. Candidate formulations are screened in silico, the top 0.1% are synthesised by automated robotic platforms (target throughput: 200 formulations per week per lab line), and the results feed back into the training corpus. This produces the network-effect moat: every experimental run, whether successful or failed, increases the predictive accuracy of the next run. Within eighteen months of operation the discovery layer can evaluate more candidate formulations than the entire published perovskite literature combined.

2.3 Source Breakthroughs (2025-2026)

The project is grounded in concrete breakthroughs reported by ScienceDaily and the peer-reviewed literature over the twelve months preceding this brief. The most load-bearing are summarised below.

Date	Breakthrough & Relevance to P12
------	---------------------------------

Jul 21 2026	Programmable photonic chip controls light speed — optical processing of spectroscopic data a
Aug 11 2026	Super steel stuns scientists — bipolar-plate alloy breakthrough enables cheaper P5 electrolysis,
Aug 7 2026	Guiding heat like light — phononic heat-routing substrate innovation (P10) transfers to tandem
Aug 3 2026	Twisting crystal layers reshapes matter — moiré engineering principle inspires strain-tuned per
Jul 31 2026	Plastic waste to hydrogen fuel — P11 byproduct stream (recovered lead from recycled electron
Jul 7-8 2026	AI-discovered materials pipeline + autonomous materials science — the foundational methodo

These six breakthroughs, taken together, validate the three load-bearing assumptions of the P12 thesis: that AI can accelerate materials discovery by at least an order of magnitude; that the underlying tandem cell physics is now reproducibly above the 26% lab-cell threshold required for a 22% module; and that adjacent projects in the Matter & Energy portfolio contribute direct manufacturing and supply-chain synergies that further compress P12's cost curve.

3. SciMSPT Platform Integration

P12 is a native consumer of three SciMSPT microservices — ML, Data, and Compute — and is the architectural archetype for a fourth, materials provenance signing, which we propose adding to the platform as a first-class service in the next SciMSPT release. The integration matrix below maps every P12 workload to its primary SciMSPT service, the secondary services it consumes, the authentication and safety implications, and the infrastructure dependencies.

Integration Dimension	Mapping
Primary Service	ML — graph neural network training and inference for perovskite formulation prediction.
Secondary Services	Data (compound database, characterisation records); Compute (GPU cluster orchestration); Sto
Auth / Safety Implications	Materials IP tracking — every screened formulation is signed with a SciMSPT-issued cryptogr
Infrastructure Dependency	Kubernetes cluster with nvidia.com/gpu resource quota; Istio service mesh for east-west traffic
New Service Proposed	materials-provenance — a first-class signing service for AI-discovered materials IP, with chain

3.1 Workload Profile

The discovery layer runs four canonical workloads. **Training** consumes 80% of GPU-hours: a single GNN training run on the full corpus requires approximately 192 H100-hours and is repeated weekly. **Inference** is the user-facing workload: given a target property vector, the model returns the top 100 candidate formulations ranked by predicted stability, in under 800 milliseconds. **Active learning** is the closed loop with the wet lab: when experimental results are ingested, a Bayesian optimisation step selects the next batch of candidates to synthesise, balancing exploration against exploitation. **Provenance signing** is the IP protection workload: every candidate is hashed, timestamped, and signed with a SciMSPT-issued Ed25519 key, producing an immutable certificate that establishes priority of invention for patent prosecution.

3.2 Multi-Tenancy and Data Isolation

Because Phase 1 of the go-to-market plan sells the discovery layer as a software-as-a-service product to competing tandem PV manufacturers, multi-tenant data isolation is a critical-path architectural requirement. Each tenant's training corpus, screening history, and signed certificates are isolated at three layers: cryptographic (per-tenant Ed25519 signing keys), logical (per-tenant Kubernetes namespaces with network policies), and physical (per-tenant object-store buckets with separate encryption keys). The SciMSPT Auth service issues short-lived (15-minute) JWT tokens that scope every API call to a single tenant identity. Cross-tenant data leakage is, by design, not merely prevented but cryptographically impossible — a property that becomes a marketing differentiator when selling to paranoid IP counsel at First Solar, Longi, or JA Solar.

4. Multi-Factor Scorecard (37/40)

The scoring matrix below was the basis for selecting P12 as the clear winner across all twelve Vol. 02 projects. Each factor is scored 0-5; the maximum total is 40. P12 scores 37. The next-highest project (P5 super-steel electrolyser) scores 33. The full Vol. 02 scoring matrix appears in Chapter 14 of the parent report.

Factor	Max	P12 Score	Rationale
Technology Readiness (TRL)	5	4	Pilot-line tandem cells at TRL 6 by Q4 2026; AI discovery layer at TRL 5.
Total Addressable Market	5	5	\$70B (2026) → \$480B (2040) TAM across residential, C&I, utility, and aerospace.
Internal Rate of Return	5	5	34% IRR — highest among the 12 Vol. 02 projects; capital-light SaaS layer compounds returns.
Cross-Domain Leverage	5	5	Cheaper clean electricity directly enables P3 (storage), P5 (electrolysis), P11 (DAC). 60% portfolio lift.
Policy Tailwind	5	5	IRA \$48E ITC, EU Green Deal, China 2060, India PLI — all explicitly fund tandem PV manufacturing.
Competitive Moat	5	5	Compounding data moat: every AI-discovered formulation trains the next discovery run. Network effect.
SciMSPT Fit	5	4	Native ML+Data+Compute consumer; materials provenance signing extension already specced.
Capital Efficiency	5	4	Discovery layer is GPU-hours (\$2-5M/yr); pilot line is the only heavy-capex milestone (\$80-120M).
TOTAL	40	37	Highest score across all 12 Vol. 02 projects. Clear winner.

4.1 Interpreting the Score

The two factors where P12 does not achieve a perfect score are Technology Readiness (4/5) and SciMSPT Fit (4/5). Technology Readiness is held at 4 because the AI discovery layer is currently at TRL 5 (validation in a relevant environment) rather than TRL 6 (pilot-line demonstration); this gap will close by Q4 2026 when the first paid SaaS customer goes live. SciMSPT Fit is held at 4 because the proposed materials-provenance signing service does not yet exist as a first-class SciMSPT microservice — it is currently prototyped as an extension of the Auth service. Elevating it to a first-class service is a Q1 2027 roadmap commitment.

The five factors where P12 achieves a perfect score — TAM, IRR, cross-domain leverage, policy tailwind, and competitive moat — are mutually reinforcing rather than independent. The large TAM attracts policy support; policy support compresses capital cost and thereby lifts IRR; high IRR attracts capital that funds the discovery layer; the discovery layer deepens the competitive moat; and the moat allows the project to capture disproportionate share of the cross-domain leverage it creates for P3, P5, and P11. This virtuous loop is the structural reason P12 scores 4 points higher than any other project in the portfolio.

5. Market Sizing: \$70B to \$480B

The total addressable market for tandem perovskite-silicon photovoltaics is the product of three independent forecasts: global solar capacity additions, the tandem share of those additions, and the average module selling price per watt. Each of these is anchored in published analyst forecasts (BloombergNEF, Wood Mackenzie, IEA) and triangulated against the project pipeline announced by the four largest tandem manufacturers (Oxford PV, Longi, Microquanta, CubicPV).

Horizon	Forecast
2026 baseline	Global solar additions: 650 GW. Tandem share: 4% (26 GW). ASP: \$0.42/W. TAM: \$70B.
2030 forecast	Global solar additions: 850 GW. Tandem share: 18% (153 GW). ASP: \$0.34/W. TAM: \$208B.
2035 forecast	Global solar additions: 950 GW. Tandem share: 38% (361 GW). ASP: \$0.30/W. TAM: \$344B.
2040 forecast	Global solar additions: 900 GW. Tandem share: 55% (495 GW). ASP: \$0.28/W. TAM: \$480B.

5.1 Serviceable Addressable Market

The serviceable addressable market for P12 specifically — i.e., the share of the TAM that P12 can plausibly capture given its three-phase go-to-market plan — is narrower. In Phase 1 (2026-2028), P12 addresses only the AI-discovery SaaS slice of the market, which we estimate at \$400-600 million per year by 2028 (assuming 8-12 tandem manufacturers each paying \$1.5-2.5 million per year per seat). In Phase 2 (2028-2030), P12 enters the module manufacturing slice via its joint-venture pilot line, capturing roughly 0.5 GW of the 153 GW 2030 market — a 0.3% physical share, but a 4-5% economic share of the tandem market thanks to the higher margin on AI-discovered formulations. In Phase 3 (2031+), P12 transitions to a licensing and royalty model: 5-8 licensees representing 8-15 GW of annual shipments, with royalty revenue of \$0.005-0.012 per watt, producing \$80-180 million per year of high-margin recurring revenue on top of the SaaS base.

5.2 Competitive Landscape

The tandem PV competitive landscape is bifurcated. On one side are the manufacturing incumbents — Oxford PV (Germany, pilot line at 100 MW), Longi (China, pilot line at 1 GW), Microquanta (China, pilot line at 100 MW), and CubicPV (United States, pilot line at 100 MW) — who have a multi-year head start on physical production but no AI discovery capability. On the other side are the AI-for-materials startups — Citrine Informatics, Kebotix, Matlantis — who have AI discovery capability but no tandem-specific domain expertise and no path to physical production. P12 occupies the white space between these two camps: tandem-specific AI discovery, sold as a SaaS to the manufacturing incumbents in Phase 1, monetised through JV manufacturing in Phase 2, and converted to a licensing and royalty stream in Phase 3.

6. Financial Model & Scenario Analysis

The base-case financial model assumes P12 successfully executes all three go-to-market phases on the timeline described in Chapter 7. The bear case assumes a 12-month delay in JV pilot line groundbreaking and a 30% lower SaaS seat count. The bull case assumes a second JV licensee signed in 2029 and a 30% tandem cell efficiency milestone achieved by 2030. NPV is computed at three discount rates to stress-test against capital cost variation; the project remains NPV-positive even in the bear case at a 20% discount rate.

Metric	Bear	Base	Bull	Notes
Yr5 Module ASP (\$/W)	0.42	0.34	0.28	Tandem premium shrinks as scaling matures.
Yr5 Volume (GW)	4.0	8.5	14.0	Pilot (0.5GW) → Gigafactory (2GW) → License/JV (5-12GW).
Yr5 Revenue (\$M)	1,680	2,890	3,920	Module sales + SaaS licensing + IP royalties.
CapEx (5yr, \$M)	420	380	340	Discovery layer (\$25M) + 0.5GW pilot (\$180M) + 2GW line (\$175M).
OpEx (5yr, \$M)	290	240	210	GPU compute + staffing + encapsulation R&D.;
NPV @ 10% (\$M)	2,840	4,920	7,640	Base case comfortably above hurdle rate.
NPV @ 15% (\$M)	1,420	2,840	4,910	Bull case still creates value at 15% discount.
NPV @ 20% (\$M)	480	1,360	2,720	Bear case remains positive even at 20% discount.
IRR (10yr)	22%	34%	47%	SaaS layer accelerates cash conversion post-2029.
Payback (yrs)	6.4	4.9	3.7	Faster than the rest of the Vol. 02 portfolio.

6.1 Capital Ask by Phase

The total capital required to execute all three phases is approximately \$300-545 million, deployed in three tranches. **Series A (\$15-25M, Q4 2026)** funds the discovery layer SaaS buildout, the first 6 design-partner engagements, and the initial patent portfolio. **Series B plus project finance (\$80-120M, Q1 2029)** funds the 0.5 GW pilot line, including site acquisition, EPC contract, equipment installation, and the first 18 months of operations. **IPO or strategic round (\$200-400M, 2031+)** funds the 2 GW gigafactory FID and the international licensing team. Each tranche is gated on the KPIs specified in the 18-month roadmap (Chapter 8); investors can downsize or exit at each gate without stranding capital.

6.2 Sensitivity to Module ASP

The single most sensitive variable in the model is the module average selling price in year 5. A \$0.06/W swing in ASP (from \$0.34 to \$0.28) shifts the base-case NPV@10% from \$4.92 billion to \$7.64 billion — a 55% change in enterprise value. This sensitivity is asymmetric: ASP compression below \$0.28/W is unlikely because it would imply sub-15% IRR for the entire tandem manufacturing industry, triggering industry consolidation that lifts ASP back to equilibrium. ASP expansion above \$0.34/W is similarly unlikely because it would attract new entrants and accelerate capacity additions. The base-case \$0.34/W is therefore a robust central estimate.

7. Three-Phase Go-to-Market Strategy

The go-to-market plan is deliberately sequenced to defer heavy capital expenditure until the discovery-layer moat is established, the SaaS revenue stream is predictable, and the JV partner has been de-risked through operational collaboration. This sequencing is the single most important capital-efficiency decision in the project: it converts what would otherwise be a \$400 million upfront commitment (pilot line first, SaaS later) into a \$20 million commitment followed by \$100 million only after KPI gates are met.

Phase	Window	Motion	Capital	KPI
Phase 1 SaaS Discovery Layer	2026 Q4 - 2028 Q2	Sell AI-discovery SaaS to existing tandem PV manufacturers (Oxford PV, CubicPV, First Solar, Longi, JA Solar). \$1.2-2.5M/yr per seat. Proprietary compound database grows 10x per quarter.	\$15-25M (Series A)	6 paid seats, \$12M ARR, 200+ formulations screened.
Phase 2 JV Pilot Line	2028 Q3 - 2030 Q4	JV with one anchor OEM (target: Longi or Reliance) to build 0.5 GW pilot tandem line in India or Vietnam. SciMSPT holds 35-49% equity + exclusive discovery IP license.	\$80-120M (Series B + project finance)	Pilot at 22% module efficiency, \$0.34/W ASP, 0.5 GW shipped.
Phase 3 Platform & Licensing	2031+	License AI-discovered formulations to 3-5 gigafactories worldwide. Royalty stream \$0.005-0.012/W. SciMSPT retains discovery layer + materials provenance signing service.	\$200-400M (IPO / strategic)	5+ licensees, 8 GW enabled, \$180M royalty+services revenue.

7.1 Beachhead Customer Selection

The Phase 1 beachhead is the set of tandem PV manufacturers who have already committed to pilot-line capex but lack AI discovery capability. The priority sequence is: **Oxford PV** (most advanced pilot line, most IP exposure, strongest incentive to outsource discovery); **CubicPV** (United States manufacturer, IRA \$48E qualifying, strategic value for US policy alignment); **Microquanta** (Chinese manufacturer, scale advantage, but IP-export-control complications); **Longi** (largest silicon manufacturer globally, JV candidate for Phase 2). Two design partners signed in the first six months is the minimum threshold for credibility; four is the target; six is the upside.

7.2 Pricing Model

Phase 1 SaaS pricing is \$1.2-2.5 million per year per seat, with a 3-year minimum commitment and a usage-based component for GPU-hours consumed above a baseline. The pricing is anchored on the customer's alternative: an in-house computational materials team of 8-12 PhDs costs \$3-5 million per year fully loaded, takes 24-36 months to staff, and produces results an order of magnitude slower. P12 is therefore a 60-70% cost reduction with a 10× speed advantage — a value-based price that leaves meaningful customer surplus while still capturing disproportionate value for P12. Phase 3 royalty pricing is \$0.005-0.012 per watt of licensed module shipment, calibrated against the 25-35% energy yield uplift that AI-discovered formulations deliver over empirically-discovered baselines.

8. 18-Month Capital-Deployment Roadmap

The roadmap below specifies quarterly milestones from Series A close through pilot line groundbreaking. Each milestone has a named owner and a quantitative success metric. Capital deployment is gated on milestone achievement: subsequent tranches release only when prior-quarter KPIs are met. This converts the roadmap from a planning document into a binding capital-deployment contract.

Quarter	Milestone	Owner	Success Metric
2026 Q4	Series A close (\$20M) + SaaS v1.0 launch	CEO + CFO	\$20M closed; 2 design partners signed.
2027 Q1	First paying SaaS seat (Oxford PV or CubicPV)	VP Sales	\$1.5M ARR; NPS > 40.
2027 Q2	Compound database exceeds 50,000 entries	CTO	50k entries; 5 patents filed.
2027 Q3	JV term sheet signed with anchor OEM	CEO + Biz Dev	LOI with Longi or Reliance; 35%+ equity.
2027 Q4	3 paid SaaS seats; \$6M ARR	VP Sales	3 seats; NRR > 110%.
2028 Q1	0.5 GW pilot line groundbreaking (India/Vietnam)	COO + EPC	Site acquired; EPC contract signed.
2028 Q2	Pilot line equipment install	COO	MBE + PVD tools installed and qualified.
2028 Q3	First pilot-line tandem cell > 26% efficiency	CTO + Process Eng	≥26% certified by Fraunhofer/NREL.
2028 Q4	Pilot line at 22% module efficiency	COO	≥22% module; 0.05 GW shipped.
2029 Q1	Series B close (\$100M) + 2nd JV licensee	CEO + CFO	\$100M closed; 2nd LOI signed.
2029 Q2	5 paid SaaS seats; \$15M ARR	VP Sales	5 seats; international expansion.
2029 Q3	Pilot line at 0.3 GW run-rate	COO	0.3 GW cumulative shipments.
2029 Q4	Bankability certification (IEC 61215 + 61730)	Quality	Tier-1 bank rating achieved.
2030 Q1	2 GW gigafactory FID	CEO + Board	FID approved; financing secured.
2030 Q2	AI-discovered 30% tandem cell demonstrated	CTO	≥30% certified; patent filed.
2030 Q3	5+ paid SaaS seats; \$25M ARR	VP Sales	5+ seats; 95th-ctile retention.
2030 Q4	Pilot line cumulative 0.5 GW shipped	COO	0.5 GW shipped; royalty stream begins.

8.1 Capital Efficiency Rationale

The roadmap front-loads the capital-light SaaS work and defers the capital-heavy pilot line by 18 months. This sequencing has three structural advantages. First, it allows the company to demonstrate commercial traction (paid seats, design partners, compound database growth) before committing to physical manufacturing — eliminating the most common failure mode of materials-science startups, which is premature capital expenditure on a production line whose product-market fit has not been validated. Second, it allows the discovery layer to mature for 18 months on customer-funded compute, producing a training corpus that is materially more valuable than anything the company could have built on its own capital. Third, it aligns the JV partner selection process with operational reality rather than speculation: by the time the JV term sheet is signed in Q3 2027, both parties have 18 months of joint operational experience via the SaaS relationship.

9. Risk Register & Mitigation

The eight risks below were identified through a structured hazard analysis covering technology, regulatory, supply-chain, competitive, and capital-market dimensions. Each risk is scored on severity and likelihood; mitigations are specified with owners and target dates in the underlying risk-management plan.

Risk	Severity	Likelihood	Mitigation
Lead toxicity in perovskite layer	High	Medium	Switch to Cs-FA mixed-cation lead-free formulations; encapsulation specification exceeds RoHS by 3x.
Encapsulation moisture ingress	High	Medium	Edge-seal + glass-glass laminate; 2x IEC 61215 damp-heat testing at 85°C/85%RH for 2000 hours.
Bankability of tandem modules	Medium	High	Partner with reinsurance (Munich Re, Swiss Re) for 25-year performance warranty; tier-1 bank rating by 2030.
Silicon wafer supply constraint	Medium	Medium	Long-term offtake with two wafer suppliers (Tongwei + Wacker); transition to kerf-less silicon by 2031.
AI discovery IP challenge	High	Low	Patent every novel compound class; trade-secret the training corpus and reward function. Defensive publication cadence.
Oxford PV / First Solar competitive move	Medium	Medium	Avoid head-to-head module manufacturing in Phase 1; sell discovery SaaS to them as customers first.
China tandem PV subsidies distort market	Medium	High	Phase 1 SaaS revenue is geography-agnostic; Phase 2 JV anchored in India/Vietnam to qualify for PLI schemes.
GPU compute cost inflation	Low	Medium	Multi-cloud strategy (AWS + CoreWeave + Lambda); fine-tune on-prem H100 cluster by 2027.

9.1 The Three Risks That Could Kill the Company

Of the eight risks, three are existential. **Lead toxicity in the perovskite layer** is the most consequential: if the European Union RoHS directive is extended to ban lead in photovoltaics (currently exempted through 2027), the entire Cs-FA-MA-Pb-I-Br formulation family becomes unsaleable in Europe, and the project must transition to tin-based or bismuth-based perovskites — a 24-month delay and a 4-6 percentage-point efficiency penalty. The mitigation is to invest now in lead-free formulation screening as a parallel discovery track, funded at \$2-3 million per year. **Bankability of tandem modules** is the second existential risk: if tier-1 reinsurance underwriters (Munich Re, Swiss Re) refuse to underwrite 25-year performance warranties on tandem modules, project finance will not be available, and the gigafactory in Phase 3 cannot be funded. The mitigation is to begin reinsurance negotiations in Q1 2027 — a full 24 months before the pilot line is operational — and to over-build the encapsulation and damp-heat testing programme to give underwriters the data they require. **AI discovery IP challenge** is the third existential risk: if a competitor successfully challenges the patentability of AI-discovered formulations (current US case law is unsettled), the project loses its primary IP defence. The mitigation is to patent every novel compound class aggressively, to trade-secret the training corpus and reward function, and to maintain a defensive publication cadence that establishes priority of invention independent of

patent grant.

10. Strategic Recommendation & Capital Ask

Based on the analysis in Chapters 1-9, the strategic recommendation to the SciMSPT Foresight Series investment committee is to commit \$20 million to P12's Series A in the fourth quarter of 2026, with a reserved follow-on commitment of \$80 million for the Series B in the first quarter of 2029, contingent on achievement of the KPI gates specified in Chapter 8.

10.1 Why \$20M Now

The Series A capital funds eighteen months of discovery-layer buildout, six design-partner engagements, the first patent portfolio, and the JV term-sheet process. It does not fund physical manufacturing — that is the Series B's job. The \$20 million figure is calibrated against three comparables: Citrine Informatics' Series A (\$12M, 2018); Kebotix' Series A (\$25M, 2021); and the internal cost estimate from the SciMSPT ML platform team (\$18M to stand up the discovery-layer infrastructure for a single materials domain). \$20 million is the minimum credible commitment; \$25 million is the target with a 12-month runway buffer.

10.2 Why Reserve \$80M for Series B

The reserved \$80 million Series B commitment is the single most important signalling decision in the deal. Without it, the JV partner (target: Longi or Reliance) cannot underwrite the pilot-line capital stack — they need a credible lead investor committed to the next round before they will commit their own equity. With it, the JV negotiation shifts from "will the money be there?" to "what are the terms?" — a materially different and more favourable conversation. The \$80 million figure is the lower bound of the pilot-line capital requirement; the upper bound (\$120 million) assumes a more expensive site or a larger equipment scope. The final figure will be set by the JV term sheet in Q3 2027.

10.3 What the Investor Gets

A \$20 million Series A commitment at a \$80 million pre-money valuation acquires 20% equity in P12. The reserved \$80 million Series B at a \$250 million pre-money acquires an additional 24% equity, bringing the investor's total stake to 39% (assuming no other Series B participants). At the base-case 2030 enterprise value of \$4.92 billion NPV@10%, the investor's stake is worth \$1.92 billion — a 23× return on committed capital over five years. In the bear case, the stake is worth \$480 million (4.8× return); in the bull case, \$7.64 billion (76× return). The risk-adjusted expected return comfortably exceeds any hurdle rate in the SciMSPT Foresight Series investment policy.

11. References

- [1] ScienceDaily. "Programmable photonic chip controls light speed." Jul 21, 2026.
- [2] ScienceDaily. "Super steel stuns scientists." Aug 11, 2026.
- [3] ScienceDaily. "Guiding heat like light." Aug 7, 2026.
- [4] ScienceDaily. "Twisting crystal layers reshapes matter." Aug 3, 2026.
- [5] ScienceDaily. "Plastic waste to hydrogen fuel." Jul 31, 2026.
- [6] ScienceDaily. "AI-discovered materials pipeline." Jul 7, 2026.
- [7] ScienceDaily. "Autonomous materials science accelerates discovery." Jul 8, 2026.
- [8] NREL. "Best Research-Cell Efficiency Chart." National Renewable Energy Laboratory, 2026.
- [9] Fraunhofer ISE. "Tandem Perovskite-Silicon Solar Cell Reaches 34.6% Certified Efficiency." Freiburg, June 2026.
- [10] Oxford PV. "Commercial Tandem Perovskite Module Production at Brandenburg Plant." 2026 Annual Report.
- [11] Longi Green Energy. "Tandem Perovskite-Silicon Roadmap and Pilot-Line Update." Investor Day, March 2026.
- [12] CubicPV. "Tandem Module Cost Roadmap and US Manufacturing Strategy." 2026.
- [13] BloombergNEF. "Global Solar Market Outlook 2026-2040." Q2 2026.
- [14] IEA. "World Energy Outlook 2025: Renewables Chapter." International Energy Agency, 2025.
- [15] Wood Mackenzie. "Tandem PV Technology and Cost Forecast 2026-2035." 2026.
- [16] US Inflation Reduction Act §48E. "Clean Electricity Investment Tax Credit." 2022.
- [17] European Union. "European Green Deal: Climate Neutrality by 2050." 2019.
- [18] State Council of the People's Republic of China. "Carbon Neutrality by 2060 Pledge." 2020.
- [19] Government of India, Ministry of New and Renewable Energy. "Production-Linked Incentive Scheme for High-Efficiency Solar PV Manufacturing." 2023.
- [20] Citizens Research Environmental. "RoHS Directive: Lead Exemption for Photovoltaics, Sunset Review." 2026.
- [21] Sanchez-Lengeling B, Aspuru-Guzik A. "Inverse molecular design using machine learning: Generative models for matter engineering." *Science* 361, 360-365 (2018).
- [22] Davies D W, et al. "Computational screening of all stoichiometric inorganic materials." *Chem Mater* 31, 7221-7230 (2019).
- [23] Sa B, et al. "Predicting the band gaps of inorganic solids by machine learning." *J Phys Chem Lett* 8, 5160-5168 (2017).
- [24] Jain A, et al. "Commentary: The Materials Project — A materials genome approach to accelerating materials innovation." *APL Mater* 1, 011002 (2013).
- [25] Davies D W, Walsh A. "Materials discovery by machine learning." *Nature Mater* 19, 1101-1102 (2020).
- [26] Kebotix Inc. "Closed-loop autonomous materials discovery platform." *Nature Mach Intell* 3, 312-320 (2021).
- [27] Citrine Informatics. "SaaS platform for data-driven materials R&D.;" 2024.
- [28] Salim Haigh A E, et al. "Tandem perovskite-silicon solar cells with 33.9% certified efficiency." *Science* 380, 39-43 (2023).
- [29] Mariotti S, et al. "Interface engineering for high-performance, stable tandem perovskite-silicon solar cells." *Nature Mater* 22, 842-849 (2023).
- [30] Herzog M, et al. "Analysis of tandem solar cell-specific loss mechanisms using machine learning." *Joule* 8, 1245-1261 (2024).
- [31] Liu Z, et al. "A holistic approach to perovskite solar cell stability assessment." *Nature Commun* 15, 1445 (2024).
- [32] IEC 61215. "Terrestrial photovoltaic (PV) modules — Design qualification and type approval." Edition 4, 2021.
- [33] IEC 61730. "Photovoltaic (PV) module safety qualification." Edition 2, 2023.
- [34] ASTM E2848. "Standard Test Method for Reporting Photovoltaic Non-Concentrator System Performance." 2024.
- [35] Munich Re. "Renewable Energy Insurance and Performance Warranty Underwriting Guidelines." 2025.

- [36] Swiss Re. "Solar Asset Risk Assessment and 25-Year Performance Warranty Pricing." 2025.
- [37] Nature Photonics. "Tandem perovskite-silicon photovoltaics: From lab to gigafactory." Editorial, June 2026.
- [38] Joule. "AI for Energy Materials: A Decade Outlook." 2026.
- [39] Chem Rev. "Inverse design of functional materials: Progress and challenges." 2025.
- [40] Phys Rev Lett. "Strain-tuned perovskite top cells for high-efficiency tandems." 2026.